



University of Glasgow | School of
Computing Science

THE AWARDS
2020

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Local Interpretable Model-Agnostic Explanations (LIME)

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Model Agnostic Approaches

- Permutation Feature Importance
- **Local Interpretable Model-agnostic Explanations**
- Shapley Additive Explanations

LIME

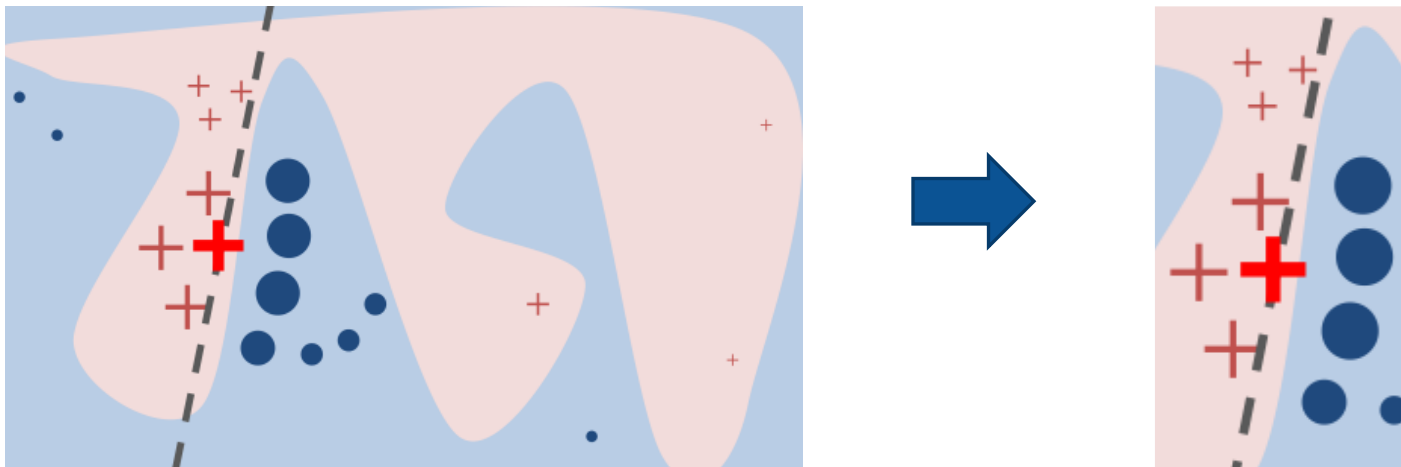
Local Interpretable Model-agnostic Explanations:

- Locally faithful explanations
- Based on a surrogate (locally linear) model

LIME

Local Interpretable Model-agnostic Explanations:

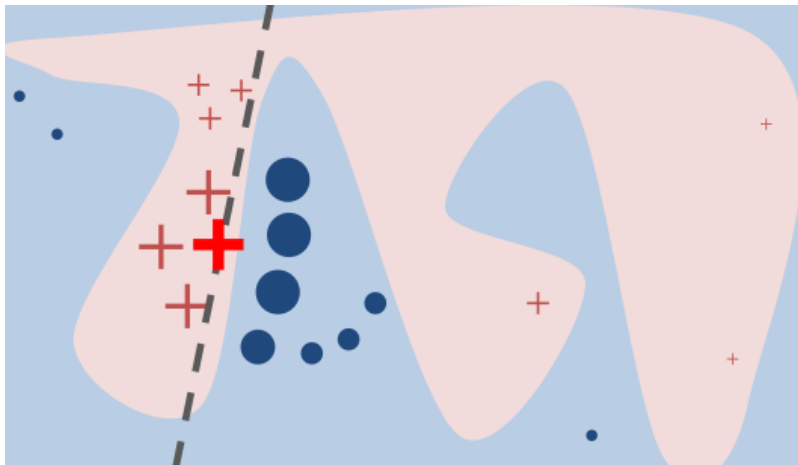
- Locally faithful explanations
- Based on a surrogate (ie. locally linear) model



LIME - Formulation

Local Interpretable Model-agnostic Explanations:

- Locally faithful explanations
- Based on a surrogate (ie. locally linear) model



Model to explain

$$f : \mathbb{R}^d \Rightarrow \mathbb{R}$$

$$\xi(x) = \operatorname{argmin}_{g \in G} \mathcal{L}(f, g, \Pi_x) + \Omega(g)$$

Explanation

Proximity
Measure

LIME – Explanations

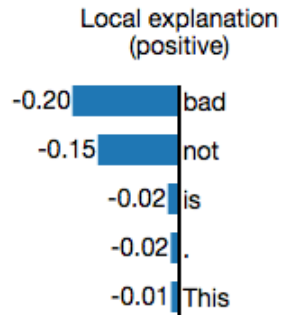
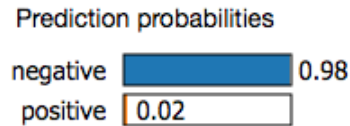
Local Interpretable Model-agnostic Explanations:

- Allow accurate explanations while it retains model flexibility
- The explanation should be accessible even to the non experts
- Small switching costs with relation to changes to the model

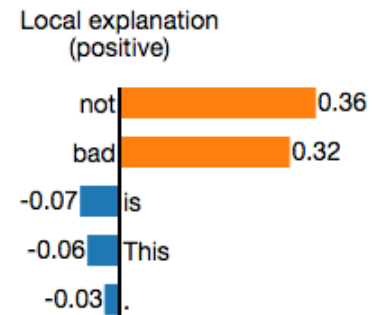
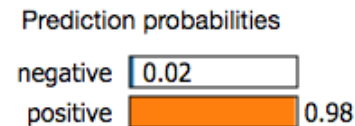
LIME – Explanations

Local Interpretable Model-agnostic Explanations:

- Allow accurate explanations while it retains model flexibility
- The explanation should be accessible even to the non experts
- Small switching costs with relation to changes to the model



This is not too bad



LIME Explanation of a logistic regression

LIME Explanation of an LSTM Model

Packaging Data for LIME

Lime Tabular
Explainer

Lime Text
Explainer

Lime Image
Explainer

Recurrent
Tabular
Explainer

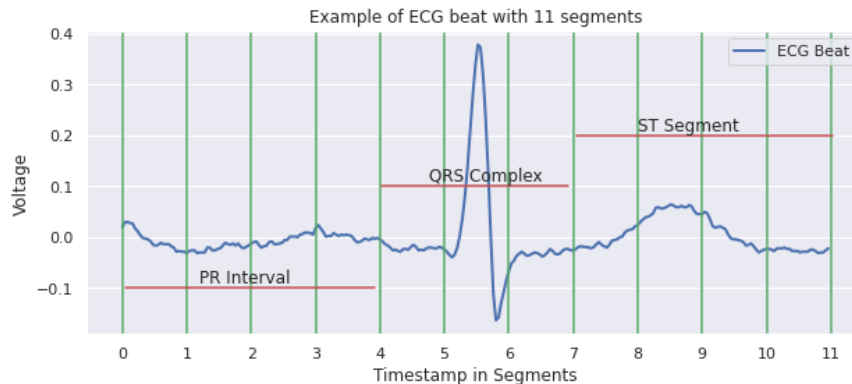
Packaging Data for LIME

Lime Tabular
Explainer

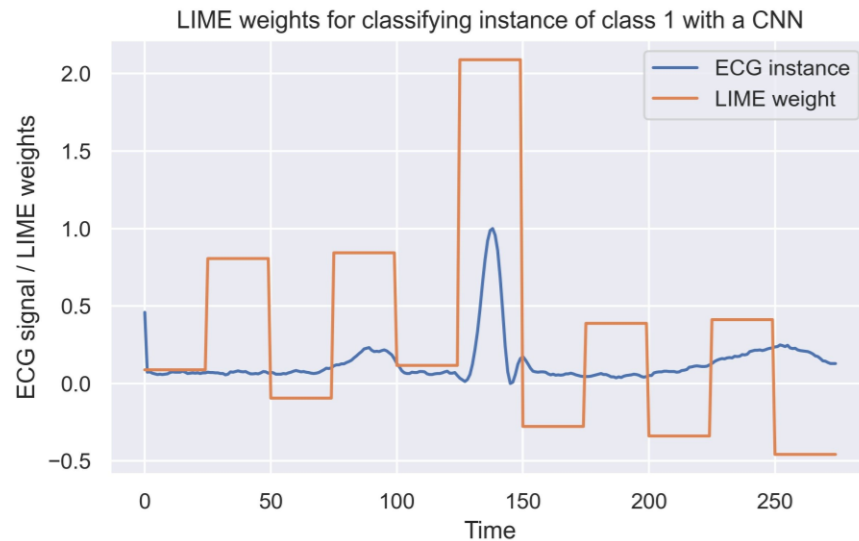
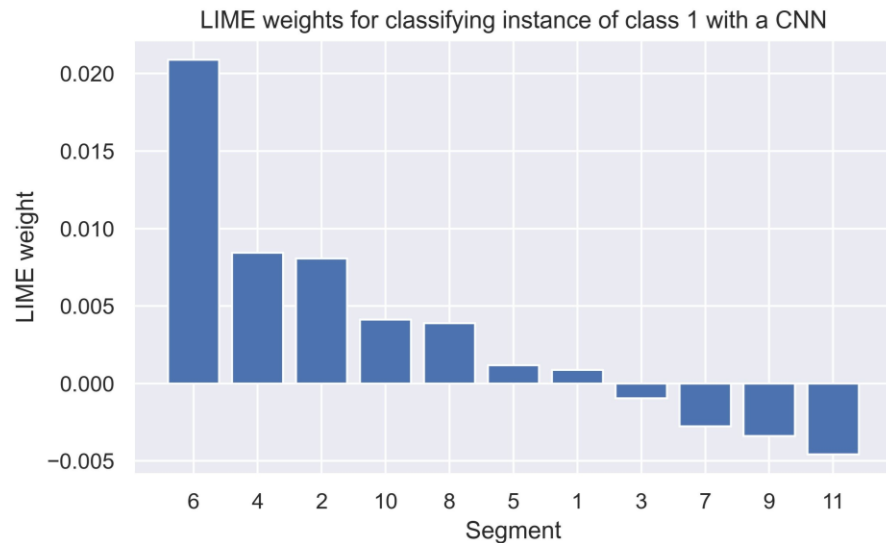
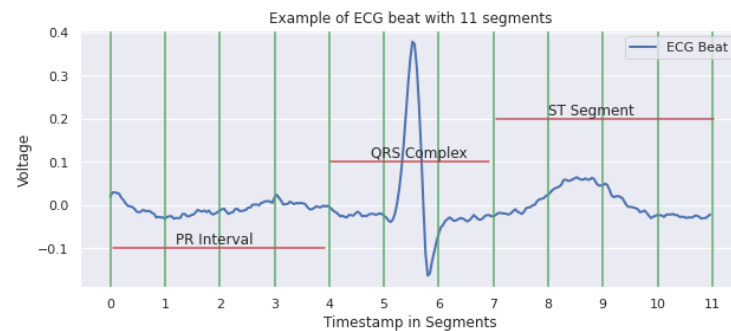
Lime Text
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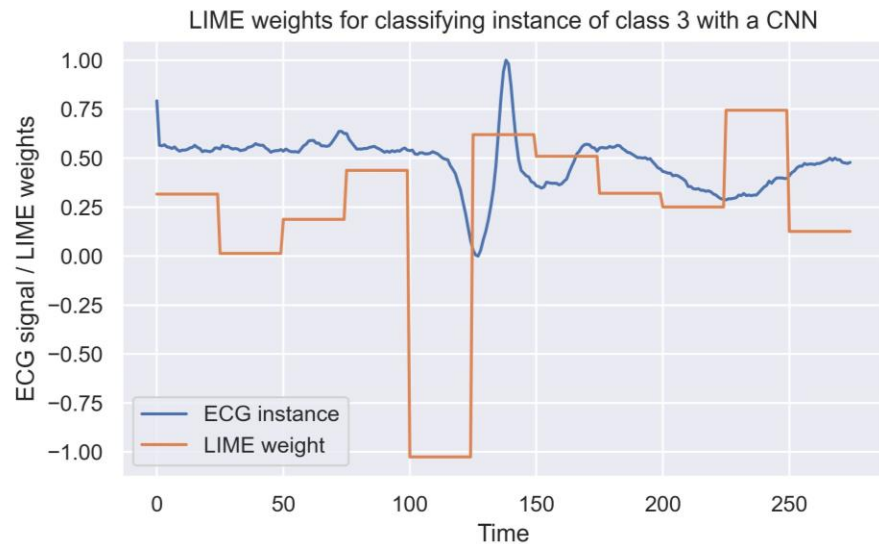
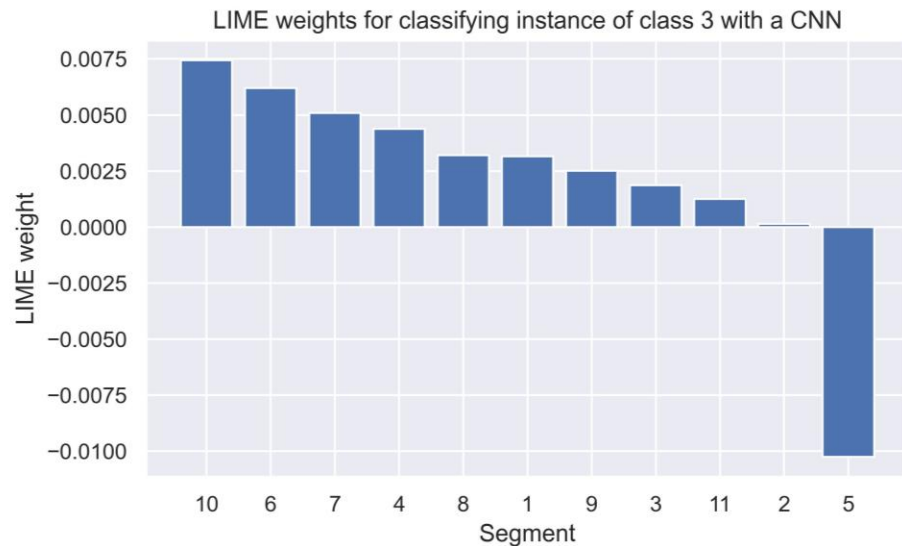
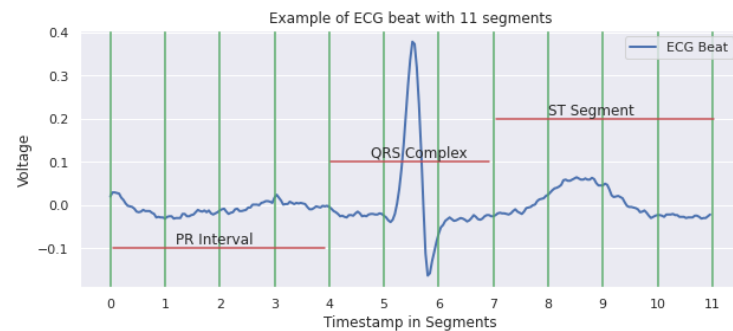
**Recurrent
Tabular
Explainer**



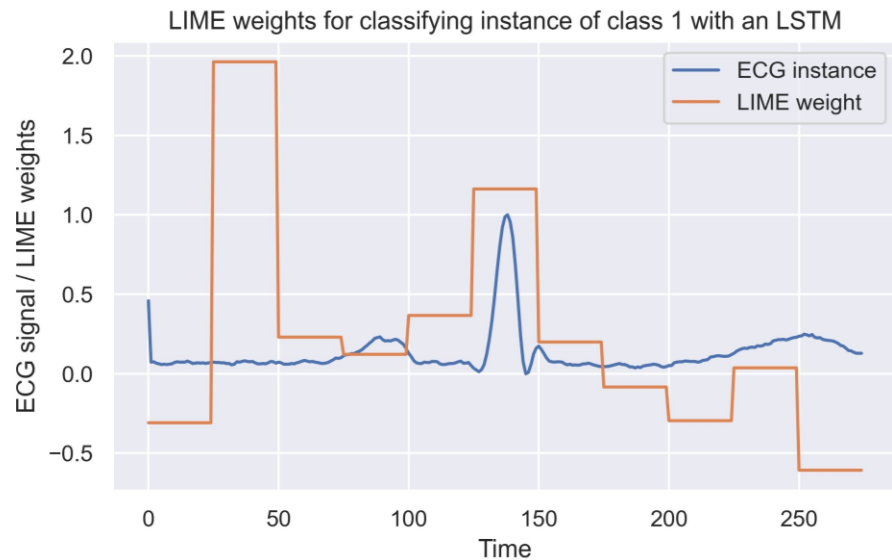
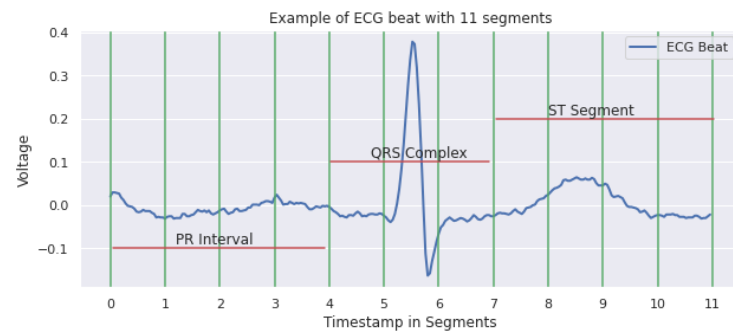
LIME for CNN Explanations



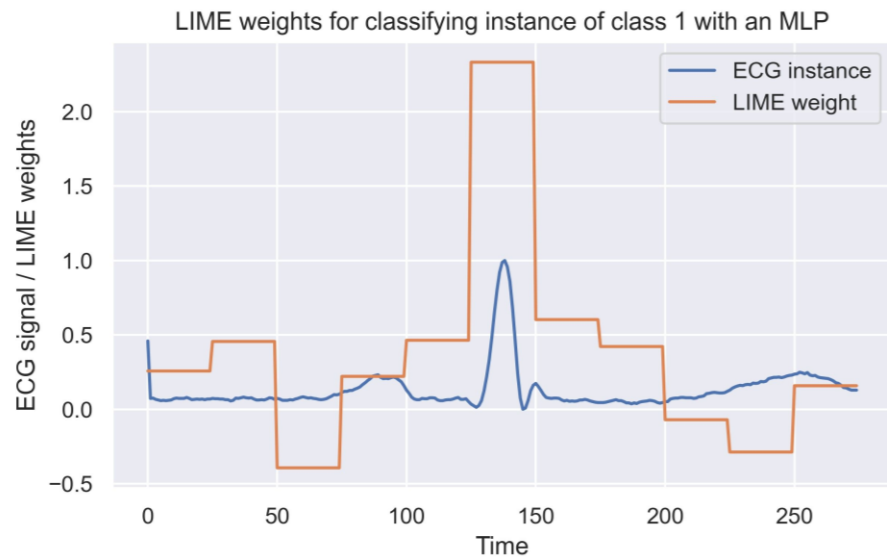
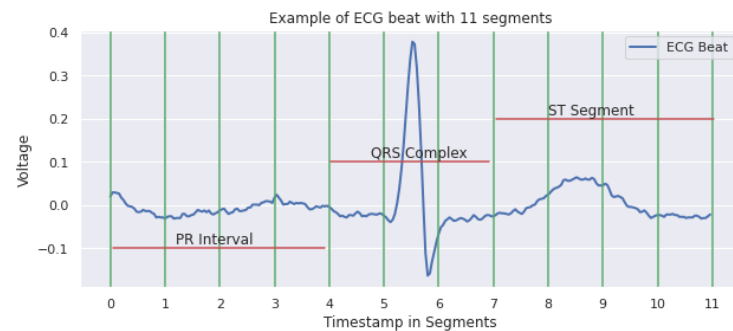
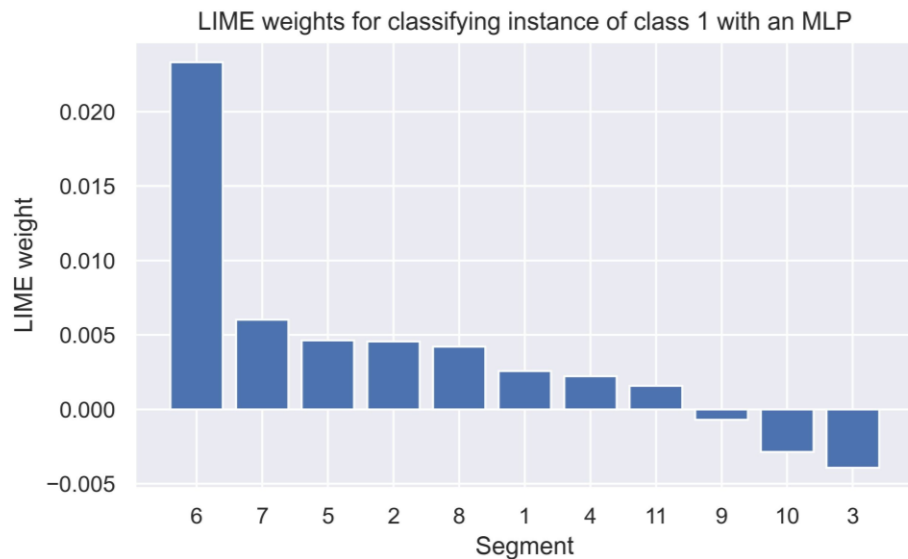
LIME for CNN Explanations



LIME for LSTM Explanations



LIME for MLP Explanations



Summary

- LIME provides a qualitative understanding between input variables and response
- LIME explanations are easily understood by non-experts
- LIME explanations are locally faithful to the model's behavior
- LIME is a model agnostic approach and thus model switching costs are small

References

- Ribeiro et al. 'Model-Agnostic Interpretability of Machine Learning', ICML Workshop on Human Interpretability in Machine Learning, 2016.
- Ribeiro et al. "Why Should I Trust You?": Explaining the Predictions of Any Classifier', Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016.



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Shapley Additive Explanations

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Model Agnostic Approaches

- Permutation Feature Importance
- Local Interpretable Model-agnostic Explanations
- **Shapley Additive Explanations**

Shapley Additive Explanations (SHAP)

- SHAP identifies a new way of estimating importance scores
- Each explanation is treated as a model itself (surrogate model)
- There is a unique solution in this class with a set of desirable properties
- Exploiting game theory guarantees a unique solution

Feature Attribution Methods

- SHAP is a member of the **additive feature attribution** methods class:

$$g(z') = \phi_0 + \sum_{i=1}^M \phi_i z'_i,$$

Kernel SHARP

- Linear LIME & Shapley values.

$$\xi(x) = \operatorname{argmin}_{g \in G} \mathcal{L}(f, g, \Pi_x) + \Omega(g)$$

Explanation **Model to explain** **Proximity Measure**

Kernel SHARP

- Linear LIME & Shapley values.

$$\xi(x) = \operatorname{argmin}_{g \in G} \mathcal{L}(f, g, \Pi_x) + \Omega(g)$$

Explanation **Model to explain** **Proximity Measure**

$$\Omega(g) = 0,$$
$$\pi_{x'}(z') = \frac{(M - 1)}{(M \text{ choose } |z'|) |z'| (M - |z'|)},$$
$$L(f, g, \pi_{x'}) = \sum_{z' \in Z} [f(h_x^{-1}(z')) - g(z')]^2 \pi_{x'}(z'),$$

Uniqueness of Additive Feature Attributions

- Local accuracy
- Missingness
- Consistency

Feature Importance - Multicollinearity

- The importance of a feature might be underestimated under the presence of multicollinearity
- Estimate every possible combination of features subsets
- The differences between the original model and every possible subset are estimated
- The total score (**shapley sampling value**) of a feature is a weighted average of all possible combinations with this feature included

Drawbacks of SHAP

- Shapley sampling values require estimating importance score for every possible subset combination of input features
- Even with KernelSHAP computational complexity is high
- With local approximators, we may still have problems to understand the model behavior
- Time-series dependencies are not taken into consideration

Summary

- SHAP is an additive feature attribution method
- SHAP handles well multicollinearity
- SHAP improves over LIME because it finds a unique solution which satisfies the properties of local accuracy, consistency and missingness

References

- Ribeiro et al. 'Model-Agnostic Interpretability of Machine Learning', ICML Workshop on Human Interpretability in Machine Learning, 2016.
- Lundberg et al. 'A Unified Approach to Interpreting Model Predictions', NIPS 2017.
- Neves et al. 'Interpretable heartbeat classification using local model-agnostic explanations on ECGs', Computers in Biology and Medicine, 2021.